

Muhammad Kashif-Khan

Email: Abdua786@outlook.com

GitHub: <https://github.com/DynoAbd786>

Portfolio Website: <https://mkkai.dev>

Personal Profile:

A Masters Computer Science and AI student at the University of Leeds, predicted to achieve a first-class honours degree, with a solid foundation in both theoretical and applied computing. My technical breadth across STEM fields, including mathematics and physics, provides a rigorous analytical framework that I apply to complex software challenges. This interdisciplinary perspective is evidenced in my project work, from designing a distributed file system to exploring AI applications in healthcare. I am seeking a graduate role in software engineering, AI, cloud, or technical consultancy where I can leverage this unique blend of deep technical skill and broad problem-solving ability.

Technical Skills:

- **Programming Languages:** Python, C/C++, C#, Java, JavaScript, SQL, Bash
- **AI & Machine Learning:** OpenCV, Scikit-learn, SpaCy NLP, JAX, Warp, Neural Networks, Computer Vision
- **Web & Cloud Technologies:** Blazor WebAssembly, .NET 8, FastAPI, Azure Functions, Cosmos DB, Redis, PostgreSQL
- **Scientific Computing & HPC:** CUDA, OpenMP, MPI, NumPy, SciPy, ParaView, CFD, XLB Framework
- **Graphics & Systems:** OpenGL, ROS2, SLAM, PyQt6, Linux/Unix, Docker, Git
- **Specialised Domains:** Medical Imaging, Healthcare AI, Distributed Systems, Biomedical Engineering

Education:

Sep 2022 – To Date MEng/BSc Computer Science with Artificial Intelligence University of Leeds

Year 2 modules:	Results:	Year 3 modules:	Results:
Data Structures and Algorithms I	84%	Robotics	86%
Software Engineering Project	81%	Parallel Computation	76%
Software Engineering Principles	78%	Computer Graphics	72%
Artificial Intelligence	75%	Machine Learning*	66%
Numerical Computation	74%	Distributed Systems*	63%

** Year 3 results affected by extended illness during semester; performance recovered in subsequent modules and projects*

Year 3 – Distributed Systems Project (Azure Serverless Inventory Management System)

- Architected serverless ERP inventory system using Azure Functions, Cosmos DB, and event-driven architecture with real-time tracking and automated stock threshold monitoring.
- Developed FaaS workflows with HTTP triggers, database change feeds, and queue integration.
- Executed performance analysis including cold start optimisation and concurrent invocation testing.
- Integrated SendGrid API for automated stock alerts with comprehensive error handling, input validation, and data integrity across distributed Azure services.

Year 2 – Data Mining Project (Medical AI with GPT-4)

- Conceptualised EPSRC-style research proposal integrating GPT-4 with medical data mining for NHS workflow optimisation in healthcare AI applications.
- Applied CRISP-DM methodology with GDPR compliance, developing explainable AI pipelines for clinical decision support and multidisciplinary healthcare integration.
- Conducted feasibility analysis for large-scale NHS deployment including cost-benefit modelling, risk assessment, and implementation strategy development.

Sep 2020 – June 2022 Allerton High Sixth Form College Leeds

- A-levels: Mathematics – **A**, Chemistry – **A**, Physics – **A**
- AS-level: Further Mathematics- **A**
- Extended Project Qualification: **A*** (Thesis – “Can We Simulate Consciousness Using AI?”)

Sep 2017 – June 2020 Allerton High School Leeds

- 11 GCSE's ranging from 9 – 6, including Maths – **9** and English - **7**

Work Experience:

June 2025 – Aug 2025 Summer Internship at the University of Leeds – Stent deployment algorithm

- Created an advanced biomedical research platform for simulating flow diverter stent deployment in patient-specific vascular geometries, extending the foundational Paliwal Virtual Stenting Workflow with modern computational methods for aneurysm treatment planning and medical device R&D.
- Engineered physics-driven stent expansion algorithms with joint-level deployment mechanics, force simulation, and collision detection systems, implementing GPU-accelerated 3D visualisation pipeline using PyVista/VTK and VMTK integration for vessel analysis and centreline extraction.
- Delivered comprehensive device repository featuring multiple enterprise stent models (Codman Enterprise, Pipeline Embolization Device, Silk Vista) with realistic wire mechanics and variable porosity modelling, establishing a foundation for clinical translation and academic research publication.

June 2024 – Aug 2024 Summer Internship at the University of Leeds – Python-based AFM Viewer

- Programmed Python-based NanoLocz platform for atomic force microscope data analysis, converting critical functionalities including video playback and mode selection using PyQt6 and parallel programming, achieving 50% performance improvement over existing MATLAB implementation.
- Designed modular, scalable codebase with OOP principles and plugin architecture supporting future extensibility, integrating advanced scientific visualisation tools (Matplotlib colour/scale bars) for enhanced data interpretation and analysis workflows.
- Led a cross-functional development team – managing project planning, task delegation, and progress tracking while collaborating with PhD researcher stakeholders to meet scientific requirements; mentored junior intern and contributed to ongoing open-source project development.

Aug 2021 – March 2022 Volunteering at Oxfam Leeds

- Independently managed the shop floor, handling sales, merchandising, and customer service, while also marketing and advertising products online through Oxfam's website.
- Coordinated in stock management of clothing, books, and toys.

June 2021 Physics work experience at The White Rose University Consortium

- Interviewed experts on *quantum computing with light*, gaining exposure to pioneering research.
- Undertook research into the fundamental principles underpinning the topic.
- Produced and presented an academic brochure to academics and researchers, clearly communicating complex ideas.

Interests and Achievements:

- Enthusiastic about exploring computer science and STEM, demonstrating curiosity and independent research skills.
- Contributed to the Formula Student AI division, applying problem-solving and software development skills to pathfinding and control of an autonomous vehicle.
- Awarded the *V100 Volunteering Award*, reflecting commitment, teamwork, and community engagement.
- Attained Bronze, Silver, Gold, and Pink Kangaroo awards in the *UK Mathematics Challenge*, evidencing strong analytical and reasoning ability.
- Provided tutoring and educational support, developing communication and mentoring skills.